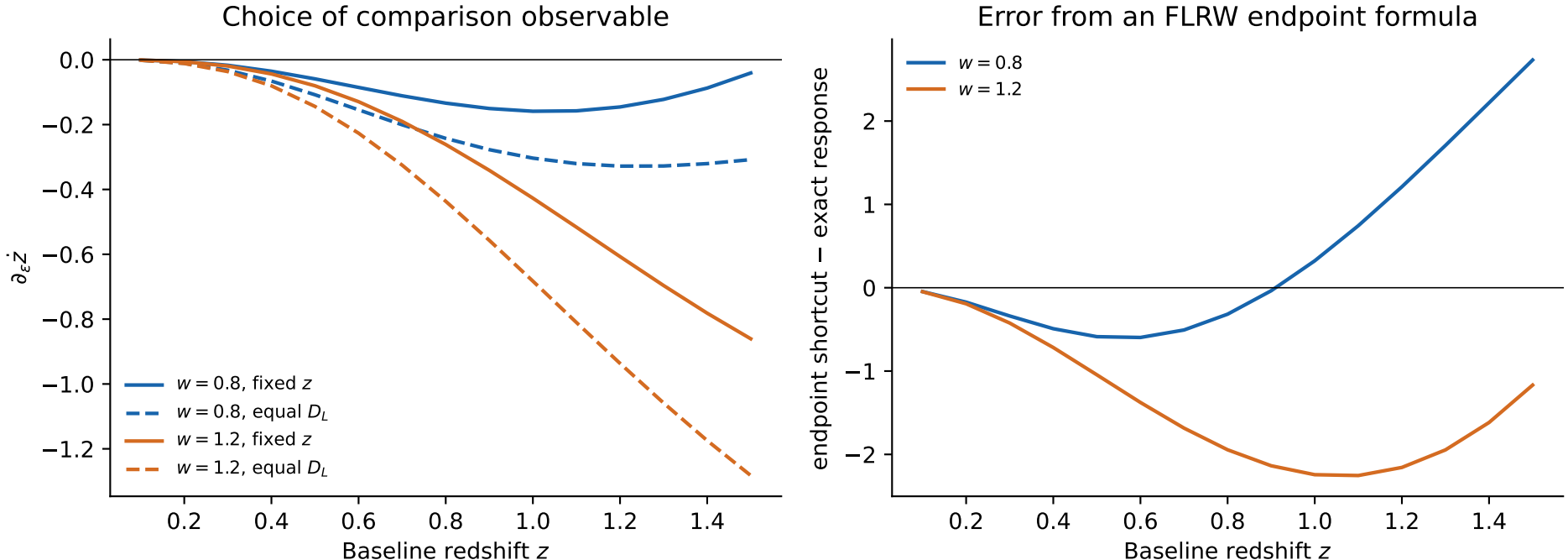
Choice of comparison observable
$\partial_\varepsilon \dot{z}$
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.2
$w = 0.8$, fixed z
$w = 0.8$, equal D_L
$w = 1.2$, fixed z
$w = 1.2$, equal D_L
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Baseline redshift z
Error from an FLRW endpoint formula
endpoint shortcut − exact response
$w = 0.8$
$w = 1.2$
2
1
0
−1
−2
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Baseline redshift z